

COLLEGE EXAMINATIONS – ACADEMIC YEAR 2020- 2021

FIRST YEAR SEMESTER II

FINAL EXAMINATION

MATHEMATICS FOR ENGINEERERS II (MAT1264)

DATE: 08/03/2021

TIME: 2 hours

MAXIMUM MARKS = 50

INSTRUCTIONS:

1. This paper contains **TWO** sections.
 2. Section A is compulsory, and Answer any **TWO** of the **THREE** questions in Section B.
 3. No written materials allowed into the Examination Room.
 4. Write all your answers in the answer booklet provided.
 5. Do not forget to write your Registration Number.
 6. Do not write any answers on this question paper.
 7. Where appropriate draw large clearly labeled diagrams in your answers.
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SECTION A (20 Marks)**Question 1.**

- (a) When a cake is removed from an oven, its temperature is measured at 350°F . Three minutes later its temperature is 200°F . How long will it take for the cake to cool off to a room temperature of 60°F ?

(3 Marks)

- b) Find the general solution of the equation $y'' + 4y' + 4y = \frac{e^{-2x}}{x^2}$.

(3 marks)

- (c) Use the operator method to find the general solution of the following system:

$$x' = 3y$$

$$y' = 2x$$

(3 Marks)

- d) If $\mathbf{f} = (2x - y^2)\mathbf{i} + (3z + x^2)\mathbf{j} + (4y - z^2)\mathbf{k}$, then find the curl of the vector $\mathbf{f}$ at the point $(1, 2, 3)$.

(3 Marks)

- e) Solve the following second order linear differential equation using Laplace transform

$$\frac{d^2x}{dt^2} + 6\frac{dx}{dt} + 9x = \sin t, \text{ with } x(0) = 0 \text{ and } x'(0) = 0.$$

(3 Marks)

- f) Evaluate $\iint x^2 y dA$ where the domain is the region $x^2 + y^2 \leq 1$.

(3 Marks)

(Hint: Use Polar coordinates).

$$r^2 \leq 1$$

- g) State the Green's Theorem in the xy plane.

(2 Marks)

SECTION B /30 Marks**Question 2:**

- a) Solve the following system of differential equations using Matrix method (eigenvalues method):

$$X' = \begin{pmatrix} -4 & 1 & 1 \\ 1 & 5 & -1 \\ 0 & 1 & -3 \end{pmatrix} X, \text{ where } X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}.$$

(8 Marks)

- b) Evaluate $\oint [2x(x+y)dx + (x^2 + xy + y^2)dy]$ around the square with vertices at $(0,0)$; $(1,0)$; $(1,1)$; and $(0,1)$.

(7 Marks)

Question 3:

Evaluate

$$\iint_R (x+y)^{-\frac{1}{2}} dA$$

over the region $x - 2y \leq 1$ and $x \geq y^2 + 1$

(7Marks)

- b) An RCL circuit connected in series has $R=10$ ohms, $C = 10^{-2}$ farad, $L = \frac{1}{2}$ henry, and an applied voltage $E=12$ volts. Assuming no initial current at $t=0$ when the voltage is first applied, find the subsequent current in the system.

(8 Marks)

Question 4:

- a) Suppose $V(x, y, z)$ volts is the electric potential at any point (x, y, z) in three-dimensional space and

$$V = \frac{1}{\sqrt{x^2 + y^2 + z^2}}$$

- (i) Find the rate of change of V at the point $(2, 2, -1)$ in the direction of the vector

$$u = 2i - 3j + 6k.$$

(5 Marks)

- (ii) Determine the direction of the greatest rate of change of V at $(2, 2, -1)$.

(3 Marks)

- b) Evaluate $\iiint_E 2x \, dv$ where E is the region under the plane $2x + 3y + z = 6$ that lies in the first

octant.

(7 Marks)

SUCCESS!!!